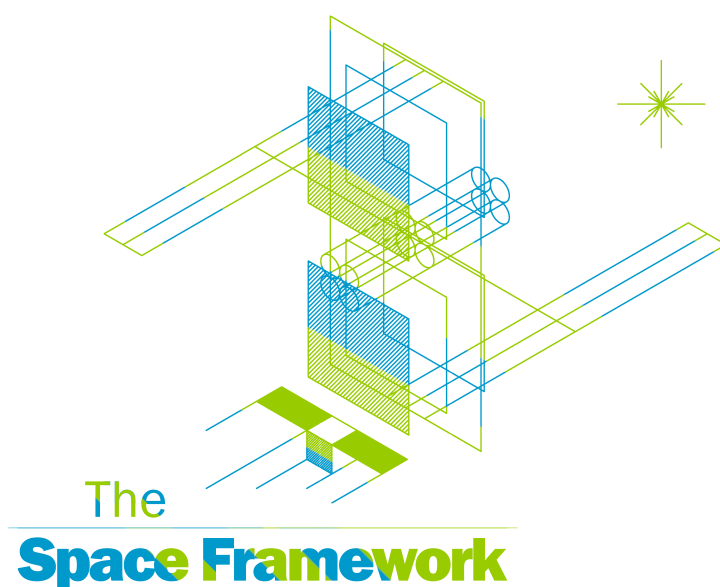


Flash Desalination for Skyraker™ ISRU Technical Demo

Complete Theoretical, Thermodynamic and Practical Guide

by Giovanni Nicola D'Aloisio



Case Study Materials (Amazon / hardware store / lab supply)

Boiler: Duran/Pyrex round-bottom flask 500 mL Amazon / lab supply

Flash chamber: Borosilicate Erlenmeyer 500 mL Amazon / lab supply

Condenser: Liebig condenser 300 mm Amazon / lab supply

Vacuum: Hand vacuum pump + gauge (0–1 bar) Amazon ≈€10

Heating: Ceramic hot plate + magnetic stirrer Lab / Amazon ≈€25

Seals: PTFE tape + silicone stoppers Ferramenta ≈€3

Tubing: Silicone tubing 6 mm ID, 1 m Amazon ≈€5

Feed: Synthetic seawater ($\text{NaCl} + \text{MgSO}_4 \cdot 7\text{H}_2\text{O} + \text{KCl}$) See §7.2

Measurement: TDS/conductivity pen Amazon ≈€8

Thermometer: K-type thermocouple + display Amazon ≈€10

March 5th, 2026

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Fact-checked: Perry's *Chemical Engineers' Handbook* 9th ed., Smith *Chemical Engineering Thermodynamics* 8th ed., ASHRAE Fundamentals 2021

Part of the Space Framework™.



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1 Overview and Motivation

Flash desalination is the process by which a pressurised liquid feed is suddenly expanded into a low-pressure chamber, causing part of the liquid to vaporise adiabatically. The vapour is subsequently condensed on a cold surface, yielding a distillate of very low salinity. The net result is:



Within the *Space Framework*, flash desalination is the first stage of the Skyraker water-processing subsystem. The distillate feeds the electrolysis unit (documented in [2]), which in turn produces the LH₂/LOX propellant for the main engine. The brine is expelled via a dedicated low-pressure valve (release mechanism TBD for orbital operation).

The choice of flash distillation over reverse osmosis (RO) is motivated by three factors:

1. **No membrane degradation.** RO membranes foul and fail; in a deep-space mission there is no replacement. A flash still has no consumable membrane.
2. **Feed-agnostic salinity.** Flash distillation is not limited by osmotic pressure: it works equally on terrestrial seawater, hypothetical European brine, or Enceladus plume material, all of which have uncertain and potentially extreme ionic compositions.
3. **Vapour integration.** The flash vapour can be partially routed to pre-heat the LH₂ prior to injection into the expander cycle, recovering enthalpy at no extra cost.

2 Thermodynamics of Flash Vaporisation

2.1 Vapour–Liquid Equilibrium and Raoult’s Law

For an ideal dilute solution (seawater approximation), the vapour pressure of water above the saline solution is depressed by the dissolved solutes according to Raoult’s law:

$$p^* = x_w p^\circ(T) \quad (2)$$

where x_w is the mole fraction of water, $p^\circ(T)$ is the saturation pressure of pure water at temperature T , and p^* is the actual vapour pressure of the solution. For standard seawater (salinity $S = 35$ g/kg):

$$x_w = \frac{n_{\text{H}_2\text{O}}}{n_{\text{H}_2\text{O}} + n_{\text{salts}}} \approx 0.9811 \quad (3)$$

$$\Delta T_{\text{BPE}} \approx 0.57^\circ\text{C} \quad (4)$$

where $\Delta T_{\text{BPE}} = T_{\text{bp,solution}} - T_{\text{bp,water}}$ is the *boiling point elevation* (BPE). This is a small but non-negligible correction in the energy balance.

2.2 Antoine Equation: Saturation Pressure of Pure Water

The saturation pressure of pure water is described by the Antoine equation (NIST parameters):

$$\log_{10}(p^\circ[\text{mmHg}]) = A - \frac{B}{C + T[^\circ\text{C}]} \quad (5)$$

Range [°C]	A	B	C	Unit
1–100	8.07131	1730.63	233.426	mmHg
60–150	8.14019	1810.94	244.485	mmHg

Table 1: Antoine constants for water (NIST, liquid phase).

Converting to SI: $p^\circ[\text{Pa}] = p^\circ[\text{mmHg}] \times 133.322$.

2.3 Flash Fraction: Isenthalpic Energy Balance & Latent Heat of Vaporisation

When a liquid at (T_f, p_f) is expanded through a flash valve to lower pressure p_{flash} , the fraction of feed that vaporises (the *flash fraction* ψ) follows from an isenthalpic energy balance across the valve:

$$h_{\text{in}} = h_{\text{out}} \implies c_{p,L}(T_f - T_{\text{flash}}) = \psi \Delta h_{\text{vap}}(T_{\text{flash}}) \quad (6)$$

Solving for ψ :

$$\psi = \frac{c_{p,L}(T_f - T_{\text{flash}})}{\Delta h_{\text{vap}}(T_{\text{flash}})} \quad (7)$$

with $c_{p,L} \approx 4.18 \frac{\text{kJ}}{\text{kg K}}$ for dilute seawater [6]. From Smith [7], steam tables linearised:

$$\Delta h_{\text{vap}}(T) \approx 2501 \frac{\text{kJ}}{\text{kg}} - 2.361 \frac{\text{kJ}}{\text{kg K}} \cdot (T[^\circ\text{C}]) \quad (8)$$

p_{flash} [mbar]	T_{flash} [$^\circ\text{C}$]	Δh_{vap} [$\frac{\text{kJ}}{\text{kg}}$]
1013	100.0	2257.0
700	90.0	2283.2
500	81.3	2304.0
300	69.1	2333.8
200	60.1	2358.3
100	45.8	2394.0
50	32.9	2423.7

Table 2: Flash temperature and latent heat as a function of pressure (pure water, NIST steam tables).

2.4 Vacuum Flash: Operating at Sub-Atmospheric Pressure

The flash temperature is set by inverting the Antoine equation at the chosen chamber pressure:

$$T_{\text{flash}} = \frac{B}{A - \log_{10}(p_{\text{flash}}[\text{mmHg}])} - C \quad (9)$$

The key advantage of vacuum flash is that the feed needs to reach only 40–70 $^\circ\text{C}$ rather than 100 $^\circ\text{C}$, reducing thermal stress on materials and heater power.

2.5 Numerical Example

Operating point: $T_f = 70^\circ\text{C}$, $p_{\text{flash}} = 200 \text{ mbar}$ ($T_{\text{flash}} = 60.1^\circ\text{C}$, $\Delta h_{\text{vap}} = 2358.3 \frac{\text{kJ}}{\text{kg}}$):

$$\psi = \frac{4.18 \times (70 - 60.1)}{2358.3} \approx 0.0175 = 1.75\% \quad (10)$$

For a feed rate of $\dot{m}_f = 100 \frac{\text{mL}}{\text{min}}$, distillate production is $\approx 1.75 \frac{\text{mL}}{\text{min}}$.

Why is ψ so small? The flash fraction is inherently 1–5% in a single-stage system. **This is not a deficiency for the laboratory demo:** the goal is to demonstrate the principle and measure ψ_{meas} vs. ψ_{theor} . Multi-stage flash (MSF) in the full Skyraker should achieve overall conversion of 20–40% by heat recovery between stages.

3 Salt Rejection and Distillate Quality

3.1 Distillate Salinity

In an ideal flash still, the vapour phase contains no dissolved salts (non-volatile solutes obey the Lewis–Randall rule: their vapour pressure contribution is negligible at these temperatures). In practice, aerosol entrainment and vapour-phase dissolution introduce a small residual content. The *salt rejection* R is defined as:

$$R = 1 - \frac{C_d}{C_f} \quad (11)$$

where C_d is the distillate conductivity [$\frac{\mu\text{S}}{\text{cm}}$] and C_f is the feed conductivity. For a well-constructed flash still: $R > 0.99$. Standard seawater: $C_f \approx 50\,000 \frac{\mu\text{S}}{\text{cm}}$. Target distillate (for electrolysis feed, ISO 22734 guidance): $C_d < 200 \frac{\mu\text{S}}{\text{cm}}$.

3.2 Brine Concentration Factor

From mass and salt balances, defining the recovery ratio $r = \dot{m}_d/\dot{m}_f$ and noting $C_d \ll C_b$:

$$C_b \approx \frac{C_f}{1 - r} \quad (12)$$

For $r = 0.0175$ (single-pass): $C_b \approx 1.018 C_f$ – negligible enrichment per pass, confirming that recirculation or multi-stage operation is required for significant water recovery.

4 Heat Transfer: Condenser Design

4.1 Overall Heat Transfer in the Liebig Condenser

The vapour condenses inside a Liebig (coaxial-tube) condenser. The heat duty is:

$$\dot{Q}_{\text{cond}} = \dot{m}_v \Delta h_{\text{vap}}(T_{\text{flash}}) \quad (13)$$

where $\dot{m}_v = \psi \dot{m}_f$. The condenser sizing follows the LMTD method:

$$\dot{Q}_{\text{cond}} = U A_c \Delta T_{\text{LM}} \quad (14)$$

$$\Delta T_{\text{LM}} = \frac{(T_{v,\text{in}} - T_{c,\text{out}}) - (T_{v,\text{out}} - T_{c,\text{in}})}{\ln\left(\frac{T_{v,\text{in}} - T_{c,\text{out}}}{T_{v,\text{out}} - T_{c,\text{in}}}\right)} \quad (15)$$

For the laboratory Liebig condenser (borosilicate glass, cooling water at 15 °C): $U \approx 500 \frac{\text{W}}{\text{m}^2\text{K}}$ (conservative estimate for condensing steam [6]).

4.2 Numerical Example: Condenser Sizing

At $\dot{m}_f = 100 \frac{\text{mL}}{\text{min}}$, $\psi = 0.0175$:

$$\dot{m}_v = 0.0175 \times \frac{100 \times 10^{-6} \frac{\text{m}^3}{\text{min}} \times 1000 \frac{\text{kg}}{\text{m}^3}}{60} = 2.92 \times 10^{-5} \frac{\text{kg}}{\text{s}} \quad (16)$$

$$\dot{Q}_{\text{cond}} = 2.92 \times 10^{-5} \times 2358.3 \times 10^3 \approx 68.9 \text{ W} \quad (17)$$

With $\Delta T_{\text{LM}} \approx 40 \text{ K}$ and a standard 300 mm Liebig condenser ($d_i = 14 \text{ mm}$, $A_c \approx 0.013 \text{ m}^2$):

$$\dot{Q}_{\text{max}} = 500 \times 0.013 \times 40 \approx 260 \text{ W} \quad (18)$$

The 300 mm condenser is adequately sized ($\dot{Q}_{\text{cond}}/\dot{Q}_{\text{max}} \approx 26\%$).

5 Materials Selection

5.1 Borosilicate Glass (Pyrex/Duran)

Borosilicate glass (SiO_2 81%, B_2O_3 13%) is chosen for all wetted components (boiler flask, flash chamber, condenser) on the basis of:

Thermal	Softening point 820 °C; safe continuous use to 500 °C; no deformation at the 40 – 80 °C operating range.
Chemical	Resistant to NaCl, MgSO_4 , KCl, and all ionic species in seawater. Does not leach contaminants into the distillate.
Visibility	Transparent, allowing direct visual observation of flash vaporisation, bubble nucleation, and distillate collection – essential for a demonstration.
Vacuum	Standard borosilicate glassware is rated for full vacuum (0 bar absolute) when free of surface defects.
Availability	Standard sizes from Amazon and AliExpress lab-ware at €3–€15 per item.

5.2 PTFE Seals and Silicone Tubing

All static seals use PTFE tape (ferramenta, €1–€2/roll); dynamic connections use silicone stoppers (chemically inert, vacuum-compatible to -0.9 bar gauge, temperature range -50 to $+200$ °C). Silicone tubing (ID 6 mm) connects flash chamber, condenser, and vacuum pump.

Why not GFRP at this stage? Glass-fibre-reinforced polymer (GFRP) with high temperature epoxy resin ($T_g > 150$ °C) is valid for the scaled-up Skyraker water treatment plant where borosilicate becomes impractical at diameters > 200 mm. For the laboratory demonstrator, borosilicate provides transparency and requires no fabrication; more information in §12.

6 Efficiency Metrics

Flash Fraction (Measured vs. Theoretical)

$$\psi_{\text{meas}} = \frac{\dot{m}_{d,\text{meas}}}{\dot{m}_f} \quad (19)$$

Gained Output Ratio (GOR) (the standard efficiency metric for desalination):

$$\text{GOR} = \frac{\dot{m}_d \Delta h_{\text{vap}}}{\dot{Q}_{\text{input}}} \quad (20)$$

For a single-stage flash still, $\text{GOR} < 1$ (typically 0.5–0.9) because the latent heat of condensation is not recovered. Multi-stage flash achieves $\text{GOR} > 8$ by heat recovery between stages.

Specific Energy Consumption (SEC)

$$\text{SEC} = \frac{\dot{Q}_{\text{input}}}{\dot{m}_d} \left[\frac{\text{kJ}}{\text{kg}} \right] \quad (21)$$

Theoretical minimum for the reference operating point:

$$\text{SEC}_{\text{min}} = \frac{\Delta h_{\text{vap}}}{\psi} \approx \frac{2358.3}{0.0175} \approx 134\,760 \frac{\text{kJ}}{\text{kg}} \quad (22)$$

Teaching point: This high SEC is expected for a single-stage demonstrator and mirrors the low $\eta_V \approx 26\%$ of the electrolysis case study: both reflect the gap between ideal and practical energy conversion at laboratory scale. In the full Skyraker MSF system, brine recirculation will reduce SEC by a factor of 5–10.

6.1 Salt Rejection

Already defined in eq. (11). The TDS pen measurement protocol is given in §10.

7 Experimental Setup

7.1 Assembly Procedure

- A1.** Inspect all glassware for chips or cracks before use. **Discard any piece with visible surface damage** – vacuum loading can cause implosion at crack tips.
- A2.** Clamp the boiler flask on the hot plate with a retort stand. Insert a one-hole silicone stopper carrying the feed inlet tube and the thermocouple wire.
- A3.** Connect the boiler outlet to the flash chamber via silicone tubing. Wrap all ground-glass joints with PTFE tape before inserting stoppers.
- A4.** Connect the flash chamber vapour outlet to the Liebig condenser inlet via silicone tubing.
- A5.** Connect the condenser outlet to the distillate collection beaker.
- A6.** Connect the vacuum pump to the flash chamber via a side port with a vacuum gauge tee.
- A7.** Seal all unused ports with PTFE-wrapped solid glass plugs or silicone stoppers.
- A8. Vacuum leak test (cold):** pump down to 200 mbar, hold for 2 min, verify pressure rises < 20 mbar. If not, tighten stoppers and repeat.

7.2 Feed Preparation

Synthetic seawater is prepared to approximate standard ocean water composition (ASTM D1141):

Salt	Mass [g]	Notes
NaCl (food grade)	23.93	Major component; supermarket
MgSO ₄ ·7H ₂ O (Epsom salt)	6.09	Divalent ions; pharmacy
KCl (potassium chloride)	0.61	Minor; pharmacy grade
Target conductivity		$\approx 50\,000 \frac{\mu\text{S}}{\text{cm}}$
Target TDS		$\approx 35 \frac{\text{g}}{\text{kg}}$

Table 3: Synthetic seawater composition for 1 L of solution.

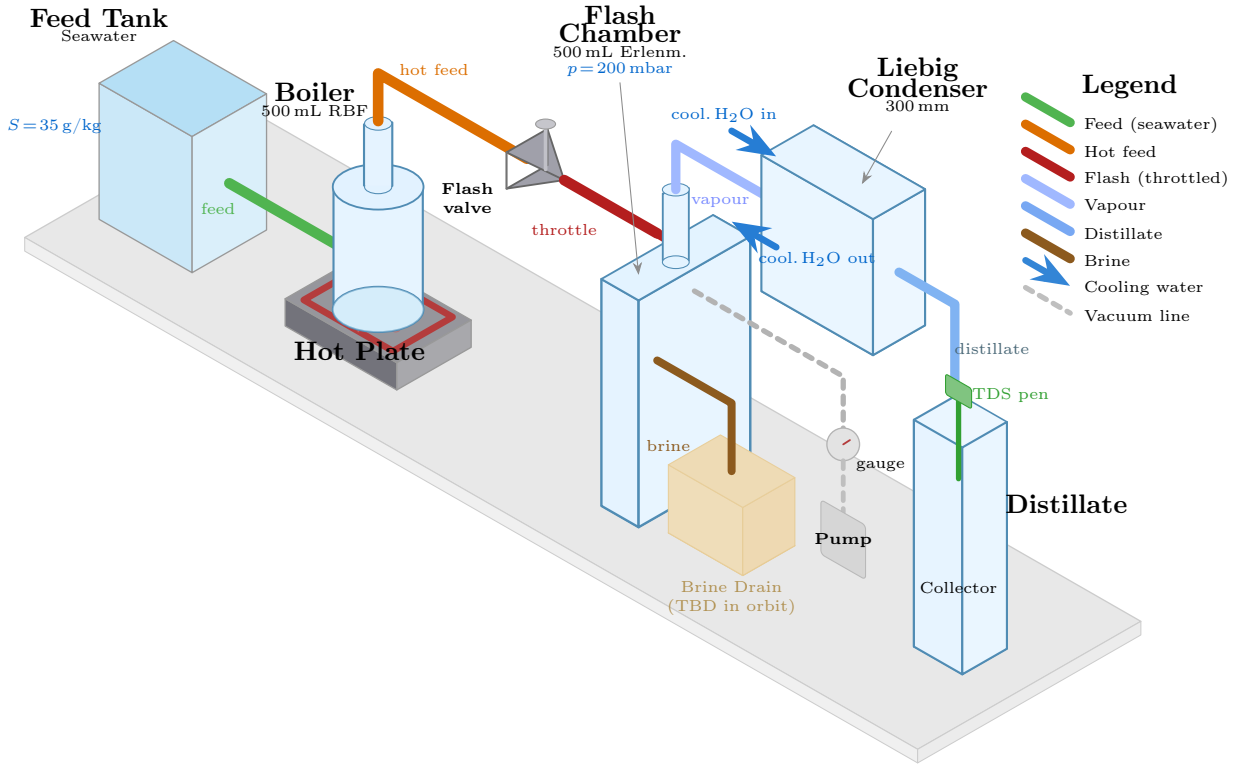
Dissolve all salts in demineralised water (same source as in the electrolysis demo [2]).

Why demineralised water? Tap water contains 10–50 mg/L Cl[−], which at the hot surface can produce HCl traces and interfere with the conductivity measurement of the distillate. Use the same demineralised water as in the electrolysis setup.

7.3 Process Flow Description & Physical Assembly Schematic

Flash Desalination – Process Flow (3D Isometric View)

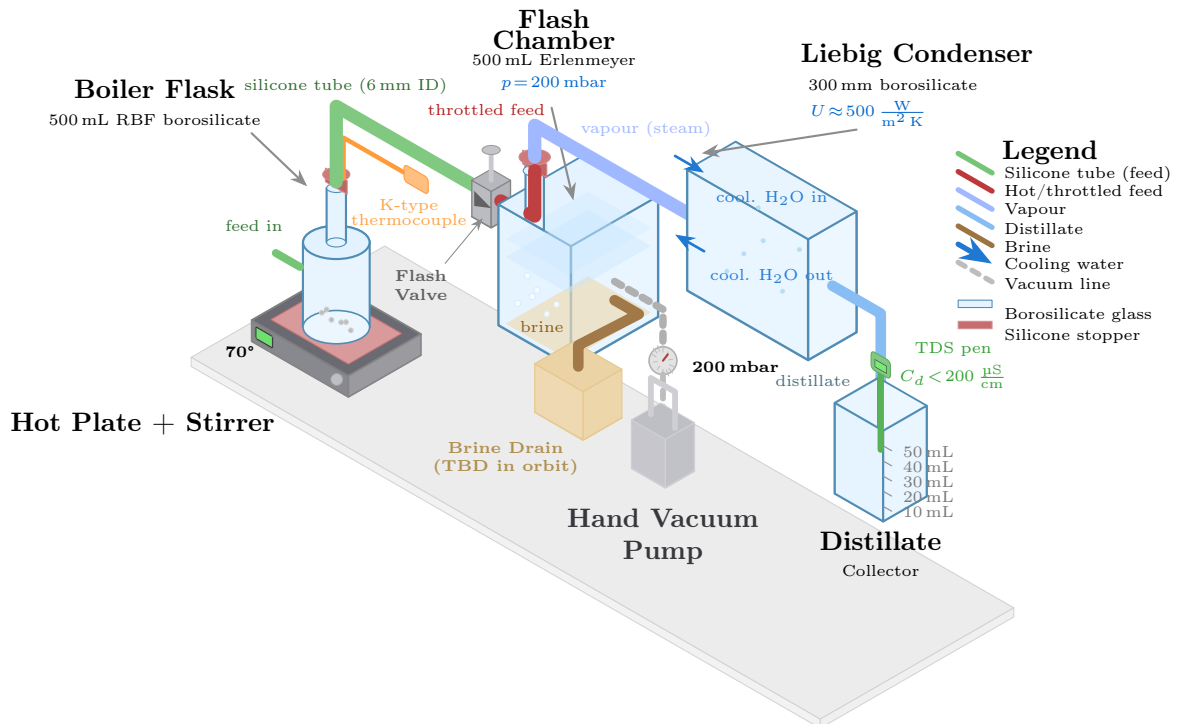
$$T_f = 70^\circ\text{C} \quad p_{\text{flash}} = 200 \text{ mbar} \quad T_{\text{flash}} = 60.1^\circ\text{C} \quad \psi_{\text{theor}} = 1.75\% \quad \dot{Q}_{\text{cond}} \approx 69 \text{ W}$$



Flash Desalination – Physical Assembly (3D Isometric View)

All wetted components: borosilicate glass

All seals: PTFE tape + silicone stoppers Cooling: tap water or ice bath



8 Case Study – Educational Demonstration

8.1 Materials List with Specifications and Purchase Links

Material	Key parameter	Technical notes	Source
RBF borosilicate 500 mL	Duran or Pyrex	Boiler. Safe to 500 °C. Vacuum-rated when crack-free. Anti-bumping granules required.	Amazon / lab supply
Erlenmeyer borosilicate 500 mL	Standard taper neck	Flash chamber. Moderate vacuum only ($p_{\text{flash}} > 100$ mbar). For lower pressures use a second RBF.	Amazon / lab supply
Liebig condenser 300 mm	$d_i \approx 14$ mm	Adequate for $\dot{Q} < 260$ W (see §4.2). Connect to tap or ice bath.	Amazon / lab supply
Hand vacuum pump + gauge	0–1 bar range	Maintains $p_{\text{flash}} = 200$ mbar. Requires periodic re-pumping ($\approx$ every 2 min).	Amazon $\approx$ €10
Ceramic hot plate + stirrer	PID control preferred	Set to 70 °C. Stirring prevents bumping. Available in most labs at no cost.	Lab / Amazon $\approx$ €25
PTFE tape 12 mm × 12 m	Food/chem grade	3–4 wraps per joint. Re-apply if any vacuum loss detected.	Ferramenta $\approx$ €1.50
Silicone stoppers (assorted)	1-hole + 2-hole + solid	Vacuum-compatible to –0.9 bar gauge. Check for cracks before use.	Amazon $\approx$ €4
Silicone tubing 6 mm ID, 1 m	Wall 1.5 mm	Compatible with vacuum and < 200 °C. Cut to length as needed.	Amazon $\approx$ €5
TDS/conductivity pen	Range 0–9990 $\frac{\mu\text{S}}{\text{cm}}$	Measure C_f (feed) and C_d (distillate) to compute R . Calibrate with known NaCl solution.	Amazon $\approx$ €8
K-type thermocouple + display	Range –50 to +300 °C	Monitor T_f in boiler. Essential to compute ψ_{theor} .	Amazon $\approx$ €10
NaCl food grade 500 g	$M = 58.44 \frac{\text{g}}{\text{mol}}$	Feed component.	Supermarket $\approx$ €0.50
MgSO ₄ ·7H ₂ O 500 g (Epsom)	$M = 246.47 \frac{\text{g}}{\text{mol}}$	Feed component.	Pharmacy $\approx$ €2
KCl 100 g	$M = 74.55 \frac{\text{g}}{\text{mol}}$	Feed component.	Pharmacy $\approx$ €1.50
Demineralised water 5 L	$\rho > 0.1 \text{ M}\Omega \text{ cm}$	Solvent for feed and for distillate conductivity reference.	Amazon / supermarket $\approx$ €1.20
Anti-bumping granules	Borosilicate or ceramic	Mandatory in boiler flask to prevent violent bumping.	Amazon / lab $\approx$ €2
Activated carbon cartridge (optional)	Inline filter	Post-distillate polishing to $C_d < 50 \frac{\mu\text{S}}{\text{cm}}$ for direct PEM electrolysis feed.	Amazon $\approx$ €2
Total budget (without lab equipment)			$\approx$ €38

Table 4: Case study materials, specifications, and purchase sources.

8.2 Working Flash Fraction Calculation

Step 1 – Set chamber pressure. Using the hand pump, bring p_{flash} to 200 mbar (readable on gauge). This sets $T_{\text{flash}} = 60.1$ °C via eq. (9).

Step 2 – Heat feed. Set hot plate to 70 °C and wait for thermocouple to stabilise within ± 1 °C.

Step 3 – Theoretical flash fraction.

$$\psi_{\text{theor}} = \frac{4.18 \times (70.0 - 60.1)}{2358.3} \approx 1.75\% \quad (23)$$

Step 4 – Theoretical distillate rate. At $\dot{m}_f = 100 \frac{\text{mL}}{\text{min}}$:

$$\dot{V}_{d,\text{theor}} = \psi_{\text{theor}} \times \dot{m}_f \approx 1.75 \frac{\text{mL}}{\text{min}} \quad (24)$$

Step 5 – Measure. Collect distillate in a graduated beaker for 600 s; measure volume $V_{d,\text{meas}}$; compute:

$$\psi_{\text{meas}} = \frac{V_{d,\text{meas}}/600}{100} \quad (25)$$

8.3 Operating Condition Sensitivity

T_f [°C]	p_{flash} [mbar]			
	50	100	200	300
50	0.73	0.18	0.00	0.00
60	1.16	0.61	0.00	0.00
70	1.56	1.01	1.75	0.00
80	1.93	1.38	0.55	0.00
90	2.31	1.75	0.93	0.12

Table 5: Flash fraction ψ [%] as a function of feed temperature and flash chamber pressure (pure water, single stage, eq. (7)).

Key insight: the dominant parameter is $\Delta T = T_f - T_{\text{flash}}$, not the absolute temperature. Lowering p_{flash} at fixed T_f is more effective than raising T_f at fixed p_{flash} , because T_{flash} drops faster than Δh_{vap} increases.

8.4 Expected Distillate Production – Data Table

Table 6: Data acquisition table. Grey columns pre-calculated; white columns filled in during experiment. ($T_f = 70$ °C, $p_{\text{flash}} = 200$ mbar, $\dot{m}_f = 100 \frac{\text{mL}}{\text{min}}$, $\psi_{\text{theor}} = 1.75\%$.)

Δt [s]	T_f [°C]	p_{flash} [mbar]	V_d^{theor} [mL]	V_d^{meas} [mL]	ψ_{meas}	C_d [$\frac{\mu\text{S}}{\text{cm}}$]
60	70.0	200	0.18			
120	70.0	200	0.35			
180	70.0	200	0.53			
300	70.0	200	0.88			
600	70.0	200	1.75			

$\psi_{\text{meas}} = (V_d^{\text{meas}}/\Delta t)/\dot{m}_f$. During the first 1–2 min, distillate rate may be lower as the flask equilibrates to the target temperature.

8.5 Efficiency and Heat Dissipation

$$\text{GOR} = \frac{\dot{m}_d \Delta h_{\text{vap}}}{\dot{Q}_{\text{input}}} \approx 0.5\text{--}0.9 \quad (\text{single stage, no heat recovery}) \quad (26)$$

$$\text{SEC} \approx 134\,760 \frac{\text{kJ}}{\text{kg}} \quad (\text{upper bound, eq. (22)}) \quad (27)$$

$$\dot{Q}_{\text{heater}} \approx 70 \text{ W} \quad (\text{hot plate at } 70 \text{ °C, } 100 \text{ mL/min}) \quad (28)$$

Why is SEC so high? At laboratory scale, single-pass operation with no heat recovery is energetically wasteful. The value is still a valid, measurable outcome: compute SEC from the wattmeter reading (if available on the hot plate) and compare with the theoretical minimum.

9 Integration with Skyraker Subsystems

The flash distillate feeds the electrolysis unit of [2]. The conductivity requirement $C_d < 200 \frac{\mu\text{S}}{\text{cm}}$ ensures compatibility with the graphite electrodes of the AWE demonstrator. An optional activated carbon cartridge (€2) inserted between the distillate collector and the electrolysis cell achieves $C_d < 50 \frac{\mu\text{S}}{\text{cm}}$ if required. The vapour stream exiting the flash chamber at $T_{\text{flash}} \approx 60^\circ\text{C}$ carries significant enthalpy. In the full Skyraker system, this vapour is routed to pre-heat the LH_2 prior to injection into the expander cycle, recovering approximately:

$$\dot{Q}_{\text{rec}} \approx \dot{m}_v c_{p,v} (T_{\text{flash}} - T_{\text{cond}}) \approx \dot{m}_v \times 1.87 \times (60 - 15) \approx 84 \dot{m}_v \quad [\text{W}] \quad (29)$$

This enthalpy recovery is included in the Phase 4 integrated model of Skyraker [1].

10 Complete Experimental Procedure

10.1 Before the Experiment

- B1. Inspect** every flask, condenser, and stopper for chips, scratches, or star cracks. **Discard any damaged piece.** A scratched round-bottom flask under vacuum can suddenly implode.
- B2. Prepare synthetic seawater.** Dissolve the salts of Table 3 in 1 L of demineralised water. Measure and record C_f with the TDS pen; target $C_f \approx 50\,000 \frac{\mu\text{S}}{\text{cm}}$.
- B3. Fill boiler.** Add 300 mL of feed plus anti-bumping granules. Clamp on hot plate.
- B4. Assemble apparatus** as per §7.1.
- B5. Vacuum leak test.** Pump to 200 mbar, hold 2 min; accept if pressure rise < 20 mbar.
- B6. Start cooling water.** Open tap or place condenser jacket inlet in ice bath.
- B7. Record ambient conditions.** Measure T_{amb} ($^\circ\text{C}$) and p_{atm} (hPa).

10.2 During the Experiment

- D1. Heat feed.** Set hot plate to 70°C . Monitor thermocouple until stable ($\pm 1^\circ\text{C}$).
- D2. Apply vacuum.** Pump down to $p_{\text{flash}} = 200$ mbar. Note time t_0 .
- D3. Open flash valve.** Remove or loosen the stopper between boiler and flash chamber. Observe flash vaporisation through the glassware.
- D4. Data logging.** Every 60 s fill in Table 6: T_f , p_{flash} (re-pump if needed), V_d^{meas} .
- D5. Conductivity.** At $t = 600$ s close flash valve, collect the distillate in a clean beaker, measure C_d *immediately* (CO_2 absorption from air raises conductivity within minutes).
- D6. Compute** ψ_{meas} , R , GOR, SEC for each row.

10.3 After the Experiment

- A1. Power off.** Stop hot plate, close vacuum valve, release vacuum *slowly* to avoid bumping.
- A2. Cool down.** Allow flask to cool to $< 40^\circ\text{C}$ before disassembly (thermal shock risk).
- A3. Drain brine.** $\text{NaCl}/\text{MgSO}_4/\text{KCl}$ solution is non-hazardous; drain with copious water.
- A4. Rinse glassware.** Rinse all components with demineralised water; dry inverted.
- A5. Post-processing.** Plot V_d^{meas} vs. Δt ; overlay V_d^{theor} ; compute ψ_{meas} , R , GOR at each point; discuss discrepancies (aerosol entrainment, vacuum stability, thermal transient).

11 Safety Regulations

Vacuum and thermal hazard profile:

Operating pressure range: 50–200 mbar (partial vacuum)

Boiler temperature: 70 °C

Steam produced: $\approx 1.75 \frac{\text{mL}}{\text{min}}$ at reference conditions

Risk of glassware implosion if cracked components are used under vacuum.

Risk of steam burns if apparatus is opened while hot.

ISO 3585:1998 Borosilicate glass 3.3 – properties.

EN ISO 3696:1995 Water for analytical laboratory use – specification and test methods.

DIN 12600 Laboratory glassware – vacuum flask requirements.

ATEX 2014/34/EU Relevant if downstream H_2 from electrolysis is generated in the same room.

Mandatory laboratory measures:

- (a) **Glassware inspection.** Discard any item with surface defects.
- (b) **Never apply vacuum to flat-bottom flasks** (Erlenmeyer used as flash chamber at moderate vacuum $p > 100$ mbar only; use a round-bottom flask for $p < 100$ mbar).
- (c) **Thermal protection.** Use insulated gloves when handling the boiler flask at 70 °C. Do not touch the condenser jacket immediately after experiment.
- (d) **Anti-bumping granules** are mandatory in the boiler flask.
- (e) **Two-person rule.** Never operate a vacuum system alone.
- (f) **Fume hood** is recommended; not mandatory at these temperatures (no toxic vapours).
- (g) **H_2 interface.** If the distillate is piped directly to the electrolysis cell in the same session, apply all hydrogen safety rules of [2] (ISO 15916:2015, ISO 22734:2019).

Perspective on risk: The dominant hazard in this experiment is *glassware implosion*, not chemical toxicity. $\text{NaCl} + \text{MgSO}_4 + \text{KCl}$ solution at these concentrations is non-toxic and non-corrosive. The 70 °C operating temperature is well within the borosilicate safe range. Rigorous pre-experiment inspection and anti-bumping granule use reduce residual risk to negligible levels.

12 Future Upgrades

Multi-stage flash A second flash stage fed by the hot brine of stage 1 increases overall recovery and GOR, as planned for the full Skyraker WTP.

GFRP pressure vessel For diameters > 200 mm, a GFRP vessel wound from woven E-glass and Araldite[®] LY1564/Aradur 2954 ($T_g \approx 155$ °C) replaces the borosilicate flask at $\approx 40\%$ mass saving.

Brine ejection valve Design and testing of a solenoid-actuated pinch valve for brine expulsion in simulated microgravity (vertical inverted test rig).

PV-powered operation Replace the lab hot plate with a PTC heater powered by the same PV array used for electrolysis, closing the energy loop for the integrated Skyraker ISRU demonstration.

13 Summary of Key Equations

Quantity	Equation	Eq.
Global process	Seawater $\rightarrow$ Distillate + Brine	(1)
Raoult's law	$p^* = x_w p^\circ(T)$	(2)
Boiling point elevation	$\Delta T_{\text{BPE}} \approx 0.57^\circ\text{C} \ (S = 35 \text{ g/kg})$	(4)
Antoine equation	$\log_{10}(p^\circ[\text{mmHg}]) = A - B/(C + T)$	(5)
Flash temperature	$T_{\text{flash}} = B/(A - \log_{10} p_{\text{flash}}) - C$	(9)
Flash fraction	$\psi = c_{p,L}(T_f - T_{\text{flash}})/\Delta h_{\text{vap}}$	(7)
Latent heat (approx.)	$\Delta h_{\text{vap}} \approx 2501 - 2.361 T \ [\frac{\text{kJ}}{\text{kg}}]$	(8)
Salt rejection	$R = 1 - C_d/C_f$	(11)
Brine concentration	$C_b \approx C_f/(1 - r)$	(12)
Condenser duty	$\dot{Q} = \dot{m}_v \Delta h_{\text{vap}}$	(13)
LMTD	$\dot{Q} = U A_c \Delta T_{\text{LM}}$	(14)
GOR	$\text{GOR} = \dot{m}_d \Delta h_{\text{vap}}/\dot{Q}_{\text{in}}$	(20)
SEC	$\text{SEC} = \dot{Q}_{\text{in}}/\dot{m}_d$	(21)
Flash fraction (measured)	$\psi_{\text{meas}} = \dot{m}_{d,\text{meas}}/\dot{m}_f$	(19)
Enthalpy recovery	$\dot{Q}_{\text{rec}} \approx 84 \dot{m}_v \text{ [W]}$	(29)

Table 7: Complete reference of key equations.

A Appendix A – ψ vs. Operating Conditions

$T_f \text{ [}^\circ\text{C]}$	$p_{\text{flash}} \text{ [mbar]}$				
	50	100	200	300	500
50	0.73	0.18	0.00	0.00	0.00
60	1.16	0.61	0.00	0.00	0.00
70	1.56	1.01	1.75	0.00	0.00
80	1.93	1.38	0.55	0.00	0.00
90	2.31	1.75	0.93	0.12	0.00

Table 8: Flash fraction ψ [%] vs. feed temperature and flash pressure. Zero entries indicate $T_f < T_{\text{flash}}$: no vaporisation.

B Appendix B – Budget Summary

Item	Source	Qty	Cost
RBF borosilicate 500 mL	Amazon / lab	1	≈€8
Erlenmeyer borosilicate 500 mL	Amazon / lab	1	≈€6
Liebig condenser 300 mm	Amazon / lab	1	≈€12
Hand vacuum pump + gauge	Amazon	1	≈€10
Ceramic hot plate + stirrer	Lab	1	€0 (lab)
PTFE tape 12 mm×12 m	Ferramenta	1	≈€1.50
Silicone stoppers (assorted)	Amazon	1 set	≈€4
Silicone tubing 6 mm ID, 1 m	Amazon	1 m	≈€5
TDS/conductivity pen	Amazon	1	≈€8
K-type thermocouple + display	Amazon	1	≈€10
NaCl food grade 500 g	Supermarket	1	≈€0.50
MgSO ₄ ·7H ₂ O 500 g	Pharmacy	1	≈€2
KCl 100 g	Pharmacy	1	≈€1.50
Demineralised water 5 L	Amazon / supermarket	1	≈€1.20
Anti-bumping granules	Amazon / lab	1 pkg	≈€2
Activated carbon cartridge (optional)	Amazon	1	≈€2
Total (without optional)			≈€71.70
Total (with optional)			≈€73.70

Table 9: Complete materials budget.

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